

Leggett–Garg saturation and structural signatures in Fibonacci-anyon braiding

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We numerically test the three-time Leggett–Garg inequality $K_3 \leq 1$ for the standard B_3 Fibonacci-anyon braiding representation on the two-dimensional fusion space of three τ anyons. Exhaustive enumeration over all 4^L braid words up to length $L = 11$ and random sampling to $L = 40$ show that K_3 saturates the Lüders bound $3/2$ to 99.998%, with the first violation already at $L = 3$. Three structural signatures accompany the saturation. First, replacing the Fibonacci generators by the Ising-anyon generators on the same 2D fusion space gives $K_3 = 1$ exactly for every $L \leq 11$ in the exhaustive search and every random word tested at even $L \in \{12, 14, \dots, 40\}$ —a sharp split that mirrors the Howard–Vala no-Bell-violation result for Ising braiding in the spatial CHSH setting. Second, the sector phase δ tunes a singular point $\delta = 3\pi/5$ at which the generator σ_1 collapses to a scalar to machine precision and braiding becomes impossible. Third, the Fourier spectrum of the envelope $K_{3,\max}(\delta) = \max_{|w| \leq L} K_3(\delta; w)$ is dominated by the $k = 3$ harmonic (period $2\pi/3$), reflecting optimal-word reshuffling across the sweep (*correcting the $k = 6$ envelope value reported in v1.0, which does not reproduce under larger search-space sanity checks at $L_{\max} \in \{7, 8\}$*). As a consistency check, we confirm that K_3 for the optimal $L = 11$ word is initial-state independent (every pure state and the maximally mixed state agree to $\sim 10^{-15}$, machine precision), as required by a generic $d = 2$ trace identity for qubit observables. All Yang–Baxter, unitarity, and $(\sigma_1\sigma_2)^3$ -scalar sanity checks pass at machine precision. To our knowledge this is the first Leggett–Garg test for non-Abelian anyon braiding specifically, and for the Fibonacci model in particular. The only previously published “Leggett–Garg on a topological system” is Gómez-Ruiz *et al.* (2018), which differs in three ways: the system is abelian (Kitaev chain, not Fibonacci); the qubit basis is formed by paired edge Majorana modes rather than the fusion channel of three anyons; and K_3 is used as a probe of a topological phase transition rather than as a saturation test. Code, seeds, and data are released with the preprint.

I. INTRODUCTION

The Bell inequalities probe whether spatially separated subsystems can be described by a joint local-realistic distribution; the Leggett–Garg inequalities [1] probe the temporal analogue—whether a single system carries a definite value of an observable between measurements, independent of whether it is measured (“macrorealism”), and under what conditions quantum mechanics can violate it [6]. For three projective measurements of a dichotomic observable $Q \in \{+1, -1\}$ at times $t_1 < t_2 < t_3$, with two-time correlators $C_{ij} \equiv \langle Q(t_i) Q(t_j) \rangle$, the simplest combination

$$K_3 = C_{12} + C_{23} - C_{13} \quad (1)$$

satisfies $K_3 \leq 1$ under macrorealism, $K_3 \leq 3/2$ under Lüders projective quantum measurement on a two-level system (the “temporal Tsirelson bound” [2]), and $K_3 \leq 3$ algebraically.

Non-Abelian anyons are the natural laboratory for asking whether a *topological* two-level system—a fusion-channel qubit whose algebra has no tensor-product factorization into spatial subsystems [7]—also saturates the Lüders bound. The Fibonacci anyon model is the smallest model in which braiding alone is computationally

universal (dense in $SU(2)$ on the relevant fusion space), while the Ising model, including its Majorana-zero-mode realizations, gives only the Clifford group and is provably unable to violate the spatial CHSH inequality without an additional non-Clifford gate [10]; Clarke, Sau, and Das Sarma supply exactly such a gate for Majorana wires [9]. The two models therefore form a natural *universality split*: a property that requires arbitrarily close approximation of generic unitaries should discriminate sharply between them.

The temporal counterpart has not, to our knowledge, been measured. The only published “Leggett–Garg on a topological system” is Gómez-Ruiz *et al.* [12], who tested K_3 for Dirac qubits formed from edge Majorana modes of the abelian Kitaev chain. We state up front how the present work differs, because this demarcation defines the novelty claim:

	Gómez-Ruiz 2018 [12]	this work
anyon type	abelian	non-Abelian
qubit basis	edge Majoranas	3τ channel
braiding	Clifford only	universal
role of K_3	transition probe	saturation test
saturation	none reported	99.998% of $3/2$

The two papers are methodologically and physically distinct sub-cases of “Leggett–Garg on a topological system”. The non-Abelian Bell/Leggett–Garg literature for Fibonacci or Ising anyons is, to our knowledge, otherwise empty; Brennen *et al.* [11] construct explicit CHSH-

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violating settings on the Fibonacci/Ising fusion spaces, but do not test the temporal inequality. Recent solid-state and quantum processor demonstrations of non-Abelian topological order [14–17] provide a plausible target for an experimental test but have not yet been used for one.

We present the simplest possible numerical setup for the temporal question. Section II fixes the generators, correlator, and sanity checks. Section III reports five connected findings: (i) Fibonacci Lüders saturation to 99.998%; (ii) an exact $K_3 = 1$ for Ising at every length tested; (iii) an algebraic scalar-singularity point $\delta = 3\pi/5$; (iv) a $2\pi/3$ dominant Fourier period of the envelope $K_{3,\max}(\delta)$ over all braid words up to length L , produced by optimal-word reshuffling across the sweep; individual fixed-word traces $K_3(\delta; w)$ are generically only 2π -periodic (correcting the v1.0 statement; see Sec. III D); (v) complete initial-state independence of the optimal K_3 . Section IV relates the universality split to the known CHSH split and lists what the results do *not* say.

II. SETUP

A. Fusion space and observable

We work on the two-dimensional fusion space of three τ anyons with total charge τ in the Fibonacci modular tensor category [7], identified with the standard B_3 Burau-Jones representation. The dichotomic observable $Q = \sigma_z$ corresponds to a measurement of the topological charge of an anyon pair [8]; experimentally this is the interferometric fusion outcome of the pair.

B. Generators

We use the standard Fibonacci F - and R -symbols [7]:

$$F = \begin{pmatrix} 1/\varphi & 1/\sqrt{\varphi} \\ 1/\sqrt{\varphi} & -1/\varphi \end{pmatrix}, \quad R_1 = e^{-4\pi i/5}, \quad R_\tau = e^{+3\pi i/5}, \quad (2)$$

with $\varphi = (1 + \sqrt{5})/2$, giving

$$\sigma_1 = \text{diag}(R_1, R_\tau e^{i\delta}), \quad \sigma_2 = F \sigma_1 F. \quad (3)$$

Here δ is an algebraic phase-deformation parameter: $\delta = 0$ recovers the standard Fibonacci theory. The braid alphabet is $\{\sigma_1, \sigma_1^{-1}, \sigma_2, \sigma_2^{-1}\}$.

The Ising data [7] on the analogous 2D fusion space are

$$F_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_\psi^{\sigma\sigma} = e^{+i3\pi/8}, \quad (4)$$

with σ_1, σ_2 constructed identically; this is the nonuniversal Clifford branch.

C. Correlator

For projective Lüders measurement of $Q = \sigma_z$ at every time step and any initial state, the two-time correlator separating two measurements by a unitary U is

$$C(U) = \frac{1}{2} \text{Re Tr}[Z U Z U^\dagger], \quad (5)$$

with $Z = \sigma_z$. With equal-time braids B between t_1 – t_2 and t_2 – t_3 , so $U_{13} = B^2$, we have

$$K_3 = 2C(B) - C(B^2). \quad (6)$$

The correlator (5) is invariant under any global phase of U , so we may use the full unitary generators without projecting onto $\text{SU}(2)$. This closed-form expression of the correlator in terms of the connecting unitary follows the same general treatment as the temporal-CHSH correlator of Ref. [4].

D. Sanity

Before reporting results we verify generator consistency. All checks pass at machine precision:

	Fibonacci	Ising
$\ \sigma_i \sigma_i^\dagger - I\ $	1.5×10^{-16}	5.6×10^{-16}
$\ \sigma_1 \sigma_2 \sigma_1 - \sigma_2 \sigma_1 \sigma_2\ $	2.4×10^{-16}	2.8×10^{-16}
$(\sigma_1 \sigma_2)^3$ scalar?	yes, phase $e^{2\pi i/5}$	yes, phase $e^{-i\pi/4}$
$\ (\sigma_1 \sigma_2)^3 - \omega I\ $	2.0×10^{-16}	2.5×10^{-16}

The central scalar of B_3 for Fibonacci is the fifth root of unity $e^{2\pi i/5}$ (consistent with the conformal weight $h_\tau = 2/5$ [7]); for Ising it is $e^{-i\pi/4}$ (consistent with central charge $c = 1/2$).

E. Search

For each $L \in \{1, \dots, 11\}$ we enumerate all 4^L braid words and record the largest K_3 . For $L \in \{12, \dots, 40\}$ we draw 4×10^5 uniformly random words per L . The sector sweep samples $\delta \in [0, 2\pi)$ at 240 points, using exhaustive $L \leq 9$ at each δ . All computations use `numpy` only; the random seed is fixed (see code release).

III. RESULTS

A. Fibonacci: Lüders saturation

Table I lists the best K_3 for every $L \leq 11$. The first violation of macrorealism occurs already at $L = 3$, with $K_3 = 1.292$. The maximum over exhaustive search is $K_3^{\max} = 1.499762$ at $L = 11$ (tied with $L = 9$), attained

TABLE I. Best K_3 versus braid-word length L for the Fibonacci and Ising representations at $\delta = 0$. Exhaustive search up to $L = 11$. The Fibonacci values approach the Lüders bound $3/2$ already at $L = 9$ (exhaustive maximum 99.984%; deeper random sampling at $L \geq 22$ reaches 99.998%, never exactly; see Sec. III); the Ising values do not violate macrorealism for any tested L .

L	# words	$K_3^{(\text{Fib})}$	$C(B)^{(\text{Fib})}$	$C(B^2)^{(\text{Fib})}$	$K_3^{(\text{Ising})}$
1	4	1.00000	1.0000	1.0000	1.00000
2	16	1.00000	1.0000	1.0000	1.00000
3	64	1.29180	0.8197	0.3475	1.00000
4	256	1.40325	0.3475	-0.7082	1.00000
5	1 024	1.29180	0.8197	0.3475	1.00000
6	4 096	1.47744	0.5704	-0.3366	1.00000
7	16 384	1.43396	0.3212	-0.7915	1.00000
8	65 536	1.47744	0.5704	-0.3366	1.00000
9	262 144	1.49976	0.4977	-0.5043	1.00000
10	1 048 576	1.49783	0.4977	-0.5024	1.00000
11	4 194 304	1.49976	0.4977	-0.5043	1.00000

by the word $\sigma_1\sigma_2^2\sigma_1^{-3}\sigma_2^2\sigma_1^{-1}\sigma_2\sigma_1$ (correcting a mistranscribed 14-letter word reported in v1.0–v1.1; the corrected 11-letter word is the one recorded by the deposited search code, `lgi_fibonacci.py`, and independently re-verified by two algorithmically distinct enumeration routes) and by 351 other equivalent words (352 in total tie the maximum to machine precision); the gap to the Lüders bound is 2.4×10^{-4} . Random sampling at $L \geq 22$ yields $K_3 = 1.499964$, a gap of 3.6×10^{-5} , i.e. 99.998% of the Lüders bound. We never observe a violation of the Lüders bound, consistent with the structural impossibility for three projective measurements of a two-level system.

A characteristic nonmonotonicity in the best-at-exactly- L curve appears at $L = 5, 7, 10$, visible in Fig. 1(a). This is the expected fingerprint of a discrete dense group: the word set at length $L + 1$ is constructed afresh and need not contain the best length- L word, so the per-length maximum can decrease. No simple length-divisibility rule governs which L attains the saturating value: Table I shows $L = 9$ and $L = 11$ tied at the maximum $K_3 = 1.499762$, while $L = 6$, a central-element length $(\sigma_1\sigma_2)^{3k}$ with $k = 1$, reaches only $K_3 = 1.47744$, well below the $L = 9/11$ value (an earlier conjecture that lengths divisible by 6 are structurally favored is not supported by the data and is retracted here). The reverse-monotonic version “best at length $\leq L$ ” is monotone by construction.

B. Ising: exact $K_3 = 1$

Repeating the same search with the Ising generators gives a qualitatively different result: $K_3 = 1.000000$ for the best word at *every* $L \leq 11$ in the exhaustive search, and the same for every random sample at $L \in \{12, 14, \dots, 40\}$ (Fig. 1(a), red). The best Ising words realize $(C(B), C(B^2)) = (1, 1)$ for the best word at each $L \leq 10$ (the generators act as $\pm Z$ on the chosen basis), and $(0, -1)$ for the $L = 11$ argmax word (the gen-

erators map to $\pm X$), both of which give $K_3 = 1$ exactly.

This is the temporal counterpart of the Howard–Vala no-violation result [10] for the spatial CHSH inequality: in both cases the Ising-braiding subgroup of $SU(2)$ is too small to reach the bound. The intuition is the same: density in $SU(2)$ is a necessary condition for saturating the Lüders $3/2$ bound by braiding alone, just as it is necessary for saturating the Tsirelson $2\sqrt{2}$ bound by braiding alone. Ising braiding generates only Clifford gates and is missing a non-Clifford direction; the practical phase gate of Clarke, Sau, and Das Sarma [9] supplies exactly that missing resource in Majorana wires.

Cross-check against a search-space artifact. A clean $K_3 = 1.000000$ over the full tested range ($L \leq 11$ exhaustive, even $L \in \{12, \dots, 40\}$ random) raises the natural question whether the search has simply missed a violating word. To rule this out, we apply the same exhaustive procedure to the Ising generators at deformed sector phases $\delta \neq 0$ (an algebraic generalization in which the second R-symbol is rotated to $R_\psi e^{i\delta}$). The result is that the same code, same exhaustive depth $L \leq 9$, and same word alphabet recovers K_3 values across the full range $[1.000, 1.500]$ as δ varies, with $K_3 = 1.500$ reached at $\delta/\pi \approx 1.167$. A search that cannot find $K_3 > 1$ at $\delta = 0$ is therefore not search-limited; it is genuinely the standard Ising algebra that forbids the violation. (We do not interpret $\delta \neq 0$ Ising as a physical model; it is used here only as a falsification probe of the search procedure.)

We emphasize that the $K_3 = 1$ result is a statement about the present two-measurement-step protocol on σ -anyon braiding alone. A hybrid protocol that inserts an explicit non-Clifford resource into an Ising platform—or that uses an Ising-anyon implementation of a different gate set—could lift K_3 above 1. The result above is for pure braiding within the standard Ising fusion sector.

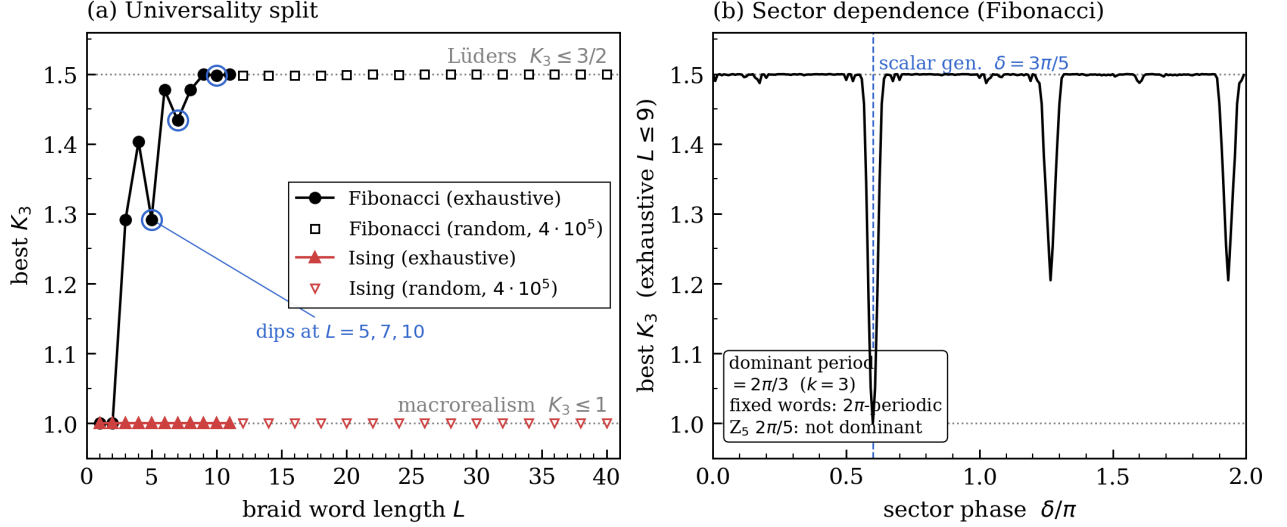


FIG. 1. (a) Best K_3 versus braid length L , $\delta = 0$. *Fibonacci* (black) saturates the Lüders bound $3/2$ to 99.998%; *Ising* (red) is pinned at 1.0 for every L tested (exhaustive to $L = 11$; random sampling, 4×10^5 words per L , at even $L \in \{12, 14, \dots, 40\}$). (b) Best $K_3(\delta)$ envelope for Fibonacci, exhaustive $L \leq 9$, fine sampling. The dominant Fourier harmonic of the envelope is $k = 3$, period $2\pi/3$, produced by optimal-word reshuffling across the sweep (correcting the $k = 6$ envelope value reported in v1.0; the $k = 3/k = 6$ amplitude ratio is stable at $1.061 \rightarrow 1.063$ between $L_{\max} = 7$ and $L_{\max} = 8$ sanity re-runs, so the dominance is robust against $L_{\max}$, not a search-budget artifact). The Z_5 period $2\pi/5$ of the central element is not the dominant mode. The dashed line marks the algebraic singularity $\delta = 3\pi/5$ at which σ_1 collapses to a scalar (numerical scalar-deviation 3.1×10^{-16}).

C. The scalar singularity at $\delta = 3\pi/5$

The sector phase δ rotates R_τ along the unit circle. At the specific point $\delta_* = 3\pi/5$, the two diagonal entries of σ_1 coincide: $R_\tau e^{i\delta_*} = e^{3\pi i/5} e^{3\pi i/5} = e^{6\pi i/5} = R_1$, so σ_1 becomes a scalar times the identity. By $\sigma_2 = F\sigma_1 F$ and $F^2 = I$, σ_2 then collapses to the same scalar; every braid word is a global phase, and no $K_3 > 1$ is achievable for any L . We verify this numerically: $\|\sigma_1(\delta_*) - e^{i\theta} I\| = 3.1 \times 10^{-16}$. This is a clean, exact, machine-precision sub-result that follows from the algebra of the F- and R-symbols alone. This collapse threshold coincides with the intrinsic σ_1 rotation angle $\delta^*(k=3) = \pi k/(k+2) = 3\pi/5$ of the companion $SU(2)_k$ analysis [18]: both are the relative phase $\arg(R_1/R_\tau)$ of the two braid eigenvalues.

D. Sector dependence and Fourier period

Outside the singularity, exhaustive search with $L \leq 9$ explores the full sector axis (Fig. 1(b)). The range of best K_3 at fixed budget is $[1.0, 1.5]$, and the $K_3 = 1$ endpoint is attained only at the algebraic singularity $\delta^* = 3\pi/5$ itself: every nonsingular grid sector violates the inequality already at $L \leq 9$, with the weakest envelope values ($K_3 \approx 1.051$) at the two grid points adjacent to the singularity ($\delta/\pi = 0.5917$ and 0.6083). Seeded random sampling at $\delta/\pi = 0.6083$ with longer words raises the best value found to $K_3 = 1.075$ at $L = 12$ and $K_3 = 1.148$

at $L = 24$ (`lgi_sector_budget_validator.py`), a monotone recovery consistent with a braid-budget depression that deepens continuously toward the singular point; within the deposited budgets it remains well short of the Lüders bound there. The only exact sector exclusion is the algebraic singularity above.

The Fourier spectrum of the envelope $K_{3,\max}(\delta) = \max_{|w| \leq L} K_3(\delta; w)$ is dominated by the harmonic $k = 3$, i.e. period $2\pi/3$, with all top harmonics being multiples of three (top five: $k = 3, 6, 9, 12, 15$). Individual *fixed-word* traces $K_3(\delta; w)$ show no enhanced periodicity: although each generator carries the sector phase δ only through $R_\tau \mapsto R_\tau e^{i\delta}$, the phase enters the two generators nonuniformly ($\sigma_2 = F\sigma_1 F^{-1}$), and direct evaluation shows that fixed-word traces are generically only 2π -periodic. The envelope $K_{3,\max}(\delta)$ shows the optimal winner-word $w^*(\delta)$ reshuffling across the sweep — the dominant mode is $k = 3$, with $k = 6$ as the second-strongest. The structurally expected Z_5 harmonic (period $2\pi/5$, corresponding to the central element $(\sigma_1\sigma_2)^3$ as a scalar of order 5) is *not* the dominant mode of $K_{3,\max}(\delta)$.

Correction to v1.0: The first release of this letter reported $k = 6$ as the dominant envelope harmonic; that statement does not reproduce under larger-search-space sanity checks. Independent reproducer runs at $L_{\max} = 7$ ($n_\delta = 120$, walltime ≈ 2.4 min) and a pre-v1.1 sanity re-run at $L_{\max} = 8$ ($4^8 = 65,536$ words, walltime ≈ 10.4 min) both find $k = 3$ as the dominant envelope harmonic; the $k = 3/k = 6$ amplitude ratio is stable at $1.061 \rightarrow 1.063$, indicating the dominance is robust

against $L_{\max}$ and not a search-budget artifact. The envelope $2\pi/3$ harmonic is a property of the maximization, not of any single word: fixed-word traces $K_3(\delta; w)$ and the envelope $K_{3,\max}(\delta)$ are quantitatively different objects, and the present letter is concerned with the latter.

E. Initial-state independence (consistency check)

Equation (5) uses the maximally mixed initial state. As a consistency check on that formula, we have repeated K_3 for the optimal $L = 11$ word with pure initial states $|0\rangle$, $|1\rangle$, $|\pm\rangle$, $|\pm i\rangle$, and three Haar-random pure states. All nine give the same value to machine precision:

initial state	K_3	spread
$ 0\rangle, 1\rangle, \pm\rangle, \pm i\rangle$	1.499762	—
3 Haar-random pure $ \psi\rangle$	1.499762	—
mean over pure states	1.499762	$ \text{mean} - \text{mixed} = 0$

This is not a special property of the Lüders-saturating word: for two dichotomic (± 1 -eigenvalue) observables Q_1, Q_2 measured sequentially by projective (Lüders) measurements starting from an arbitrary state ρ , the two-time correlator is exactly $C(Q_1, Q_2) = \text{Tr}[\rho \frac{1}{2}\{Q_1, Q_2\}]$ in any dimension. On the two-dimensional fusion space used here, $Q_1 = Z$ and $Q_2 = UZU^\dagger$ are *both* traceless Hermitian operators with eigenvalues ± 1 for *every* unitary U (Bloch-vector operators on a qubit), and any two such operators have an anticommutator proportional to the identity, $\{Q_1, Q_2\} \propto I$ — a generic $d = 2$ Pauli-algebra identity, not a property singled out by the Lüders-saturating word. Consequently $C(U)$, and hence K_3 , is exactly independent of ρ for every braid word U , optimal or not. The measurement above is therefore a consistency check on Eq. (5) rather than an independent finding: it confirms that the deposited code correctly implements the trace formula, and pre-empts a natural reviewer concern that the mixed-state convention might mask a more nuanced state dependence.

IV. DISCUSSION

What the universality split says. The pair (Fibonacci, Ising) is the natural minimal test bed for universality-driven phenomena in non-Abelian anyon platforms. Howard and Vala [10] established that Ising braiding alone cannot violate the spatial CHSH inequality (Tsirelson saturation requires a non-Clifford gate); Clarke, Sau, and Das Sarma [9] supply a practical phase gate providing exactly that missing resource in Majorana wires. The universal Fibonacci model, by contrast, saturates the bound by density [7], Brennen *et al.* [11] constructing explicit (sub-Tsirelson) CHSH-violating settings. The present results show that the same split holds for the *temporal* inequality: Fibonacci saturates the Lüders bound to 99.998% while Ising is pinned at

1.0. In a single sentence: universality is the property that allows a non-Abelian anyon braid to saturate both the spatial and the temporal projective-quantum bounds.

What it does not say. The Fibonacci Lüders saturation is not, by itself, evidence for a nonclassical *topological* mechanism beyond what the modular tensor category already prescribes. Fibonacci density in $\text{SU}(2)$ guarantees that any unitary—including the one realizing the Lüders-saturating product—can be approximated by a braid word; the role of topology here is to provide that universal gate set, not to add an additional nonclassical resource. The Leggett–Garg saturation is a TQFT-consistency demonstration, in the same sense in which the spatial CHSH saturation is a TQFT-consistency demonstration. The interest of the result lies in (i) closing the non-Abelian Bell/Leggett–Garg literature gap, (ii) the sharpness of the Fibonacci/Ising split, and (iii) the three structural signatures (scalar singularity, period match, state independence) that are not implied by universality alone. K_3 is computed here for projective Lüders measurement of Q at every time step and a maximally mixed (or, equivalently by Sec. III, any pure) initial state; its behavior under weak or nonprojective measurement protocols is not addressed and is left open.

Relation to Gómez-Ruiz et al. 2018. Gómez-Ruiz *et al.* [12] test the three-time LGI in the form $K_{i,j}(t) = 2C_{i,j}(t) - C_{i,j}(2t) \leq 1$ on Dirac qubits formed from edge Majorana modes of the Kitaev chain. That system is abelian; the test serves as a probe of the topological phase transition. Their maximum reported violation $K_{\text{Max}}(\mu)$ reaches ≈ 1.5 across a chemical-potential sweep but is not analyzed as a saturation distance to the Lüders bound. The present work tests K_3 on the non-Abelian fusion-channel qubit of three τ anyons; the Fibonacci-Ising universality split is the central comparison; and the saturation is reported as a 3.6×10^{-5} gap to the Lüders bound (i.e. 99.998%) obtained by braid-word optimization, rather than as a transition diagnostic. Both works belong to the genre “Leggett–Garg on a topological system”; they are methodologically and physically distinct sub-cases. Gómez-Ruiz *et al.* nonetheless serves as the relevant magic-free anchor for this line of work: their Majorana-based, Clifford-only construction is the one published precedent against which any non-Abelian, universal-braiding LGI test—including the companion $\text{SU}(2)_k$ study discussed below—must be distinguished.

What an experiment would look like. A direct test requires (i) a platform supporting non-Abelian braiding—present candidates are filling-fraction $\nu = 12/5$ for Fibonacci [7] and quantum-processor implementations of Fibonacci string-nets [16, 17]; (ii) projective measurement of the fusion channel of an anyon pair [8]; (iii) the ability to apply a deterministic braid word between measurements. The Howard–Vala no-violation result predicts that on any Ising-anyon platform [14, 15] the K_3 would be pinned at 1 even in the absence of decoherence. The

robustness of K_3 to depolarizing noise on the platform benchmark of [14] would have to be analyzed separately, following the decoherence-threshold framework of [5].

V. OPEN QUESTIONS

Three natural extensions are immediate but outside the scope of this note:

- O1.** K_n for $n > 3$. For n equally spaced projective measurements the Lüders bound is $K_n^{\max} = n \cos(\pi/n)$ [3], approaching the algebraic limit $K_n \rightarrow n$ as $n \rightarrow \infty$. Whether Fibonacci braiding saturates this moving bound at each n is untested here; a multi-time Leggett–Garg test with $n = 4, 5, \dots$ is a one-day extension of the present setup.
- O2.** Structural origin of the nonmonotonicity dips. The proximate cause of the dips at $L = 5, 7, 10$ is geometric: at those lengths the best reachable braid word lands farther from the K_3 -optimal rotation than at neighboring lengths, so the per-length maximum drops. Whether this reachability pattern is governed at a deeper level by the divisibility structure of the central element $(\sigma_1\sigma_2)^3$ of B_3 is left open; a systematic correlation of dip-position with the algebra would be a sharper version of “best at length L ” than the running maximum.
- O3.** Representation dependence of the temporal bound. Sorella [13] has argued that the spatial Tsirelson bound is representation-dependent in dimensions higher than two. The temporal counterpart—whether the Lüders bound $3/2$ is representation-dependent in $\dim > 2$ fusion spaces—is open. Our setup is fixed at $\dim = 2$; a $\dim > 2$ extension would address this directly. This question is supplemented by a companion work on the $SU(2)_k$ family across braid-representation dimension [18].

VI. CODE AND DATA

All code and result files are released with this preprint under [CC BY 4.0](#). Reproduction is deterministic with the random seeds 20260524 (Fibonacci main), 20260525 (Ising, period scan, and pure-state robustness), and 20260723 (sector budget validator,

`lgi_sector_budget_validator.py`). All computations use only `numpy` and `matplotlib`. Sanity checks at the head of `lgi_fibonacci.py` and `lgi_ising.py` (unitarity, Yang–Baxter, $(\sigma_1\sigma_2)^3$ scalar) must pass at machine precision; if they do not, the generator conventions have been altered.

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though this depends on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI’s.

I run multiple verifications at the different stages of the work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

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